

Export: query, preview, progress, and cancellation

Status: Working. Export is a separate subtab and runs read-only queries under the connected Salesforce identity.

User and data flow

- **1. Target:** Object choices are limited to queryable objects returned by global describe.
- **2. Fields:** Users select readable described fields. Compound address/location containers are omitted; their concrete readable fields remain selectable where exposed.
- **3. Filter:** One structured field/operator/value predicate is supported. Object and field identifiers are allowlisted by describe and validated; string values are SOQL-escaped, numeric and Boolean values are type-checked. Raw SOQL is not accepted.
- **4. Preview:** A LIMIT 20 query displays records before a full export.
- **5. Export:** Query API pages are followed through nextRecordsUrl. Progress reports loaded versus total rows. The current implementation caps one export at 50,000 rows and stops early when known daily API use exceeds 90 percent.
- **6. Cancel:** Cancel aborts in-flight requests. Partial results are deliberately not offered as a download, and history records that the run was cancelled.
- **7. Download:** Completed rows are escaped into CSV locally and downloaded. The generated dataset remains in memory only.

Salesforce APIs, permissions, and record access

Concern	Implementation
Object and field access	Global/object describe supplies queryable objects and the connected user-visible field set.
Record access	REST Query API executes in the connected user context, so sharing, role hierarchy, teams/manual shares, restriction rules, and other platform access controls determine returned rows.
Pagination	GET /query starts the request; nextRecordsUrl is followed page by page by queryAll.
API protection	SalesforceClient caps concurrent requests, retries idempotent GETs on 502/503/504, tracks Sforce-Limit-Info, refreshes once after 401, and refuses cross-instance credential use.

Errors, privacy, and operational limits

- SOQL construction errors surface before the request; Salesforce and network errors use the shared error explanation UI.
- Export history is org-scoped aggregate metadata only. Record values, previews, and generated CSV are not written to extension storage.
- The current filter builder supports one predicate and six comparison operators. Multi-condition groups, Bulk API 2.0 jobs, resume-after-restart, and exports above 50,000 rows are **Planned / not implemented**.
- The import path uses REST record operations in batches of 25, not Bulk API 2.0. Server-side bulk job monitoring and restart persistence are **Planned / not implemented**.

Traceability

Tests: tests/dataTransfer.test.ts. UI: DataImport.tsx and DataExport.tsx. Core helpers: dataTransfer.ts. Client method: SalesforceClient.upsertRecord. Navigation: routes.ts and App.tsx. This addendum reflects the repository after implementation and supersedes any earlier statement in this guide that omits the Data Import & Export section.